



Analysis of Simple Electrical Impedance Tomography Using Conditional Models

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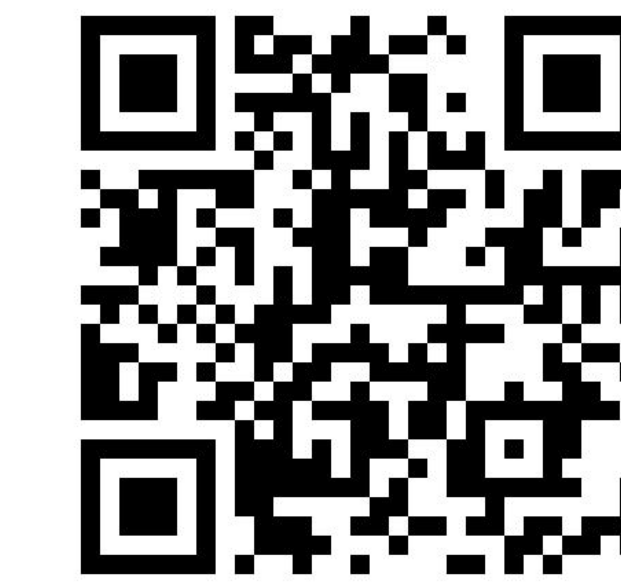
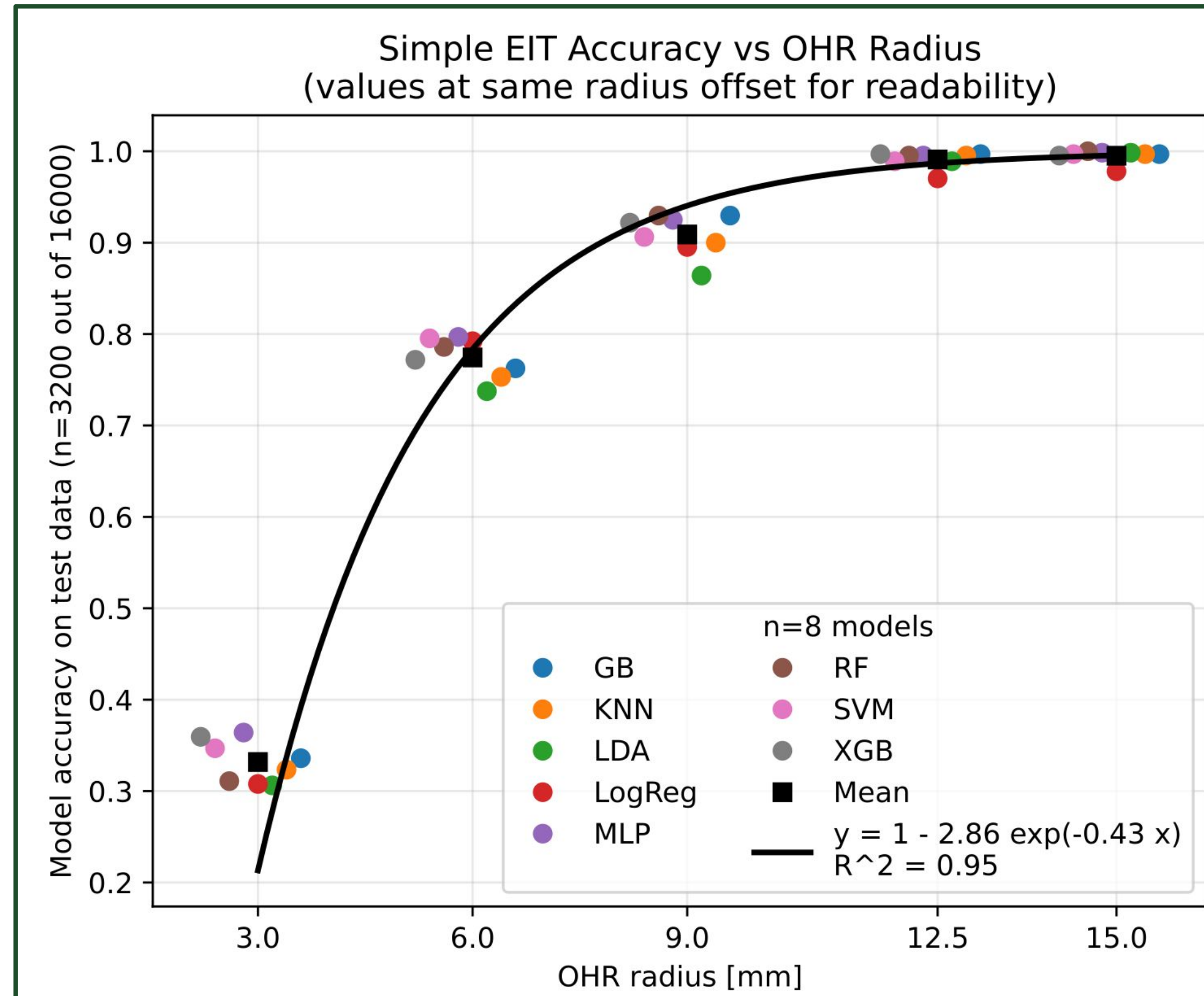
Background, Introduction, Research Question

Electrical Impedance Tomography (EIT) is a medical imaging technique that visualizes the internal resistivity of the human body, often used for real-time monitoring of the lungs. This method involves measuring voltages across multiple electrodes while applying alternating current through others. This generates a non-linear, ill-posed boundary value problem that, when solved, produces a 2D admittance tomogram. Here we present a simplified 4-electrode EIT device that utilizes empirical algorithms to estimate the confidence of detecting high-resistivity objects in one of 16 sectors. Our approach visualizes the results as a circle of sixteen sectors, each corresponding to a physical area within a water tank. Each sector indicates the confidence level that the object of higher resistivity (OHR) is present in that region. **We evaluate the accuracy of conditional models on 5 plastic PLA cylinders of radii: 3.0, 6.0, 8.0, 12.5, and 15.0 mm (curc_e to a) to determine how the radius of the OHR effects Simple EIT accuracy.**

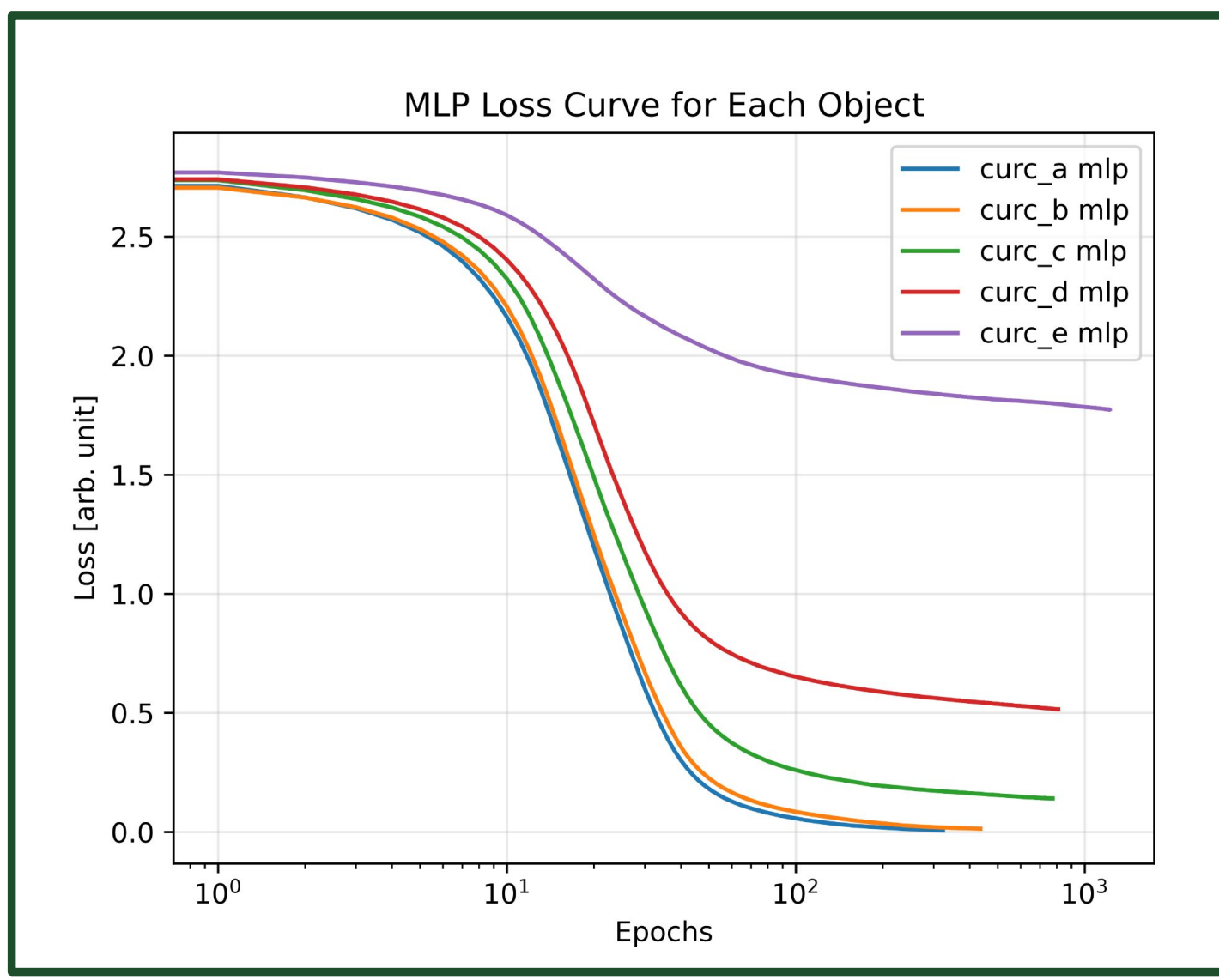
Hardware and Software

More details, including actual code, extended documentation, 3D model files, ML model files, and CSV data on GitHub. Hardware: Raspberry Pi 3B, two MUX36D04EVM-PDK, EDUX1002G Keysight oscilloscope, EDU33212A Keysight waveform generator, banana hook/plug connectors, 3D printed PETG test rig and PLA OHRs, stainless steel plates, alligator clips. Python 3 external libraries: gpiozero, joblib, matplotlib, numpy, pandas, pyvisa, scikit-learn, scipy, xgboost. The figures below show actual model output, the experimental setup, and the full circuit diagram.

Results and Data



<https://github.com/ihstas0/simple-eit>



Data Collection Methods

We initialized the wavegen parameters to 2.5 V peak-to-peak and 2.5 V offset; the oscilloscope to 0.2 ms/div. and 10 V/div., with AC coupling to remove DC offset; the MUX DC power supply to 10 V for Vdd. 100 mL of tap water was used in the test rig. To record instrumentation data we took n=1000 measurements with no OHR at various frequencies and salinities using an automatic script (recording one measurement every 0.5s). Next, for each OHR, we collected data at each of the 16 sectors, n=200 per sector, with frequency = 10 kHz in tap water. We moved the OHR around each sector while the automatic script ran to capture data for each position in the sector³. Total measurements: n=24000. The number of voltages measured: v=144000. Each measurement took 6 voltages, one for each of the 6 configurations.

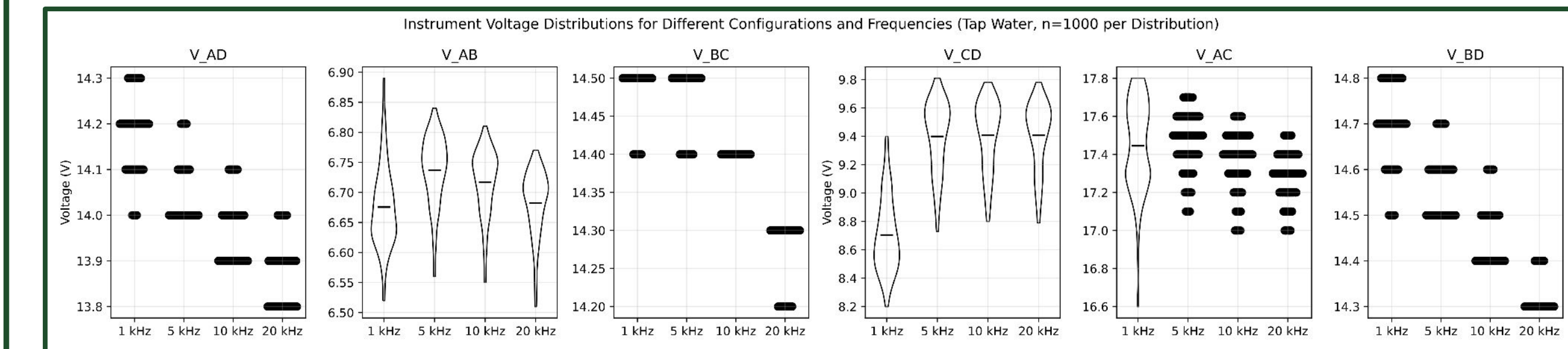


Figure D3: Instrument voltage data in tap water for frequencies for 6 voltage configurations

Conditional Model Training, Data Analysis

We gathered 1.6 mb of data. We trained 8 models on these data, with parameters:

```
xgb → XGBClassifier(n_estimators=200, learning_rate=0.05,
                    max_depth=4, verbosity=0)
gb → GradientBoostingClassifier()
knn → KNeighborsClassifier(n_neighbors=6)
lda → LinearDiscriminantAnalysis()
logreg → LogisticRegression(max_iter=2500)
mlp → MLPClassifier(hidden_layer_sizes=(10, 10), max_iter=2500)
rf → RandomForestClassifier(n_estimators=200)
svm → SVC(kernel="rbf", probability=True)
```

The conditional models, intended to assign probabilities to each of the classes (sectors) given tabular voltages, and their parameters are arbitrary for the purpose of this experiment as the assignability of the dataset to the classes itself is under question. Model accuracies for each object are plotted to the left, with loss curves.

Conclusions

1) Figure R1 shows a clear exponential relationship between object radius and model accuracy. For example, a model accuracy of 30% implies that the corresponding voltage distributions are separable 30% of the time. At 0% accuracy, the distributions are completely indistinguishable, meaning the presence of an object produces no detectable change. Therefore, extrapolating to the point where accuracy reaches 0% suggests that **Simple EIT cannot detect objects with a radius smaller than approximately 2.5 mm in real-time classification.**

2) This research shows the importance of feature separability for conditional model performance, as overlapping distributions reduce accuracy. Although distributions for smaller objects overlap, their means remain distinct. This suggests that, with a large sample size to approach the mean, even small objects can be detected.

3) In the context of EIT, it is clear that relatively high resolution is possible with few electrodes when the dimensions of the object are known and the medium static, which could, although unlikely, have applications for finding objects in known anatomy using simple conditional models. This may present a major research opportunity.

No analysis was made on conditional model overfitting, but this is likely negligible due to model choice and the resulting accuracies of different models aligning with one another, although further analysis is needed. LDA, KNN, LogReg, and GB models often underperform other models and their exclusion from the data may improve R².

Instrument reliability, especially for recording RMS voltage quickly with consumer hardware, was poor and the data would benefit from more robust hardware.

³Further discussion on the experimental impact of moving the OHR while recording data is necessary, although it is unlikely this movement would effect the conclusions.

Acknowledgements

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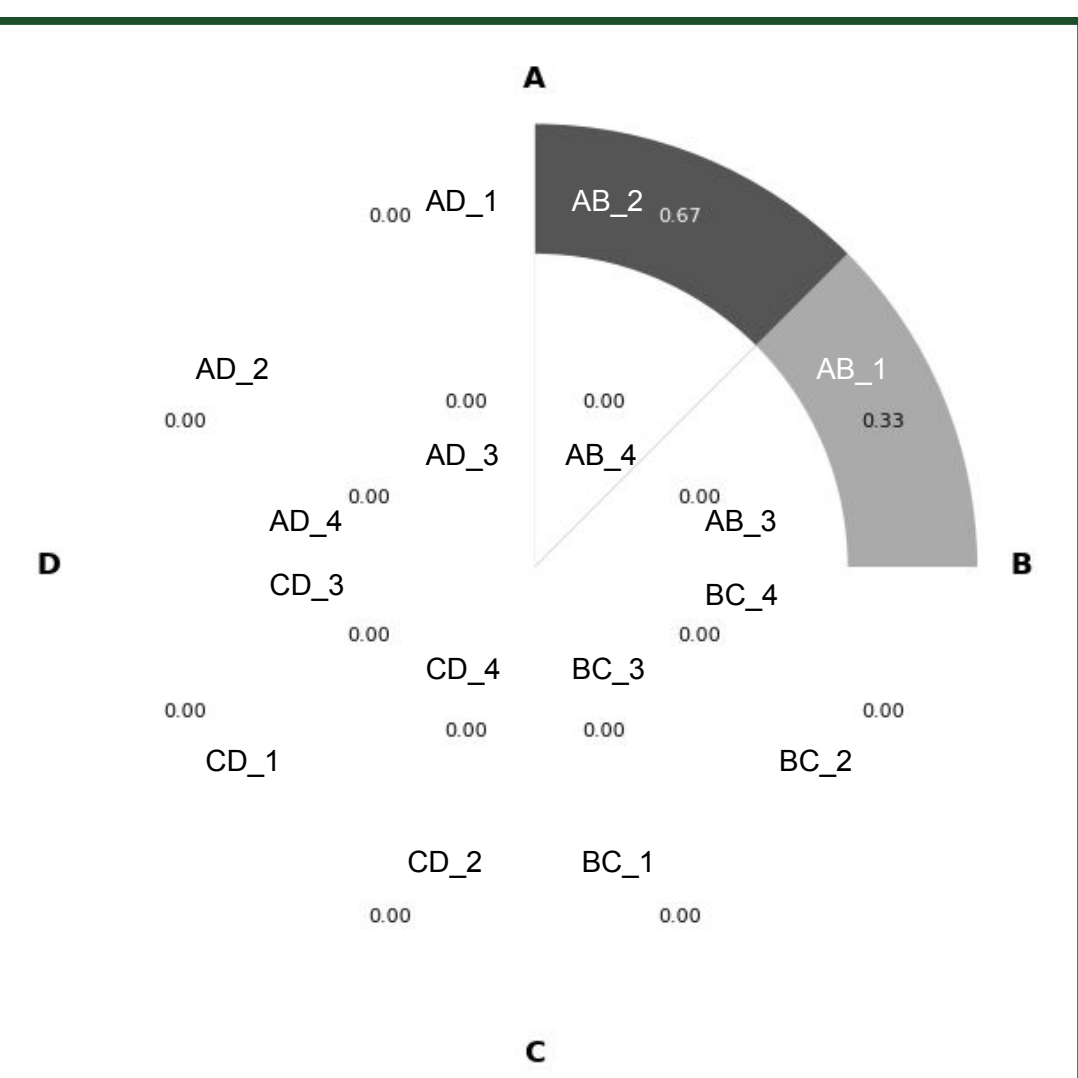


Figure H1: Sector visualization (0.67 at AB_2)

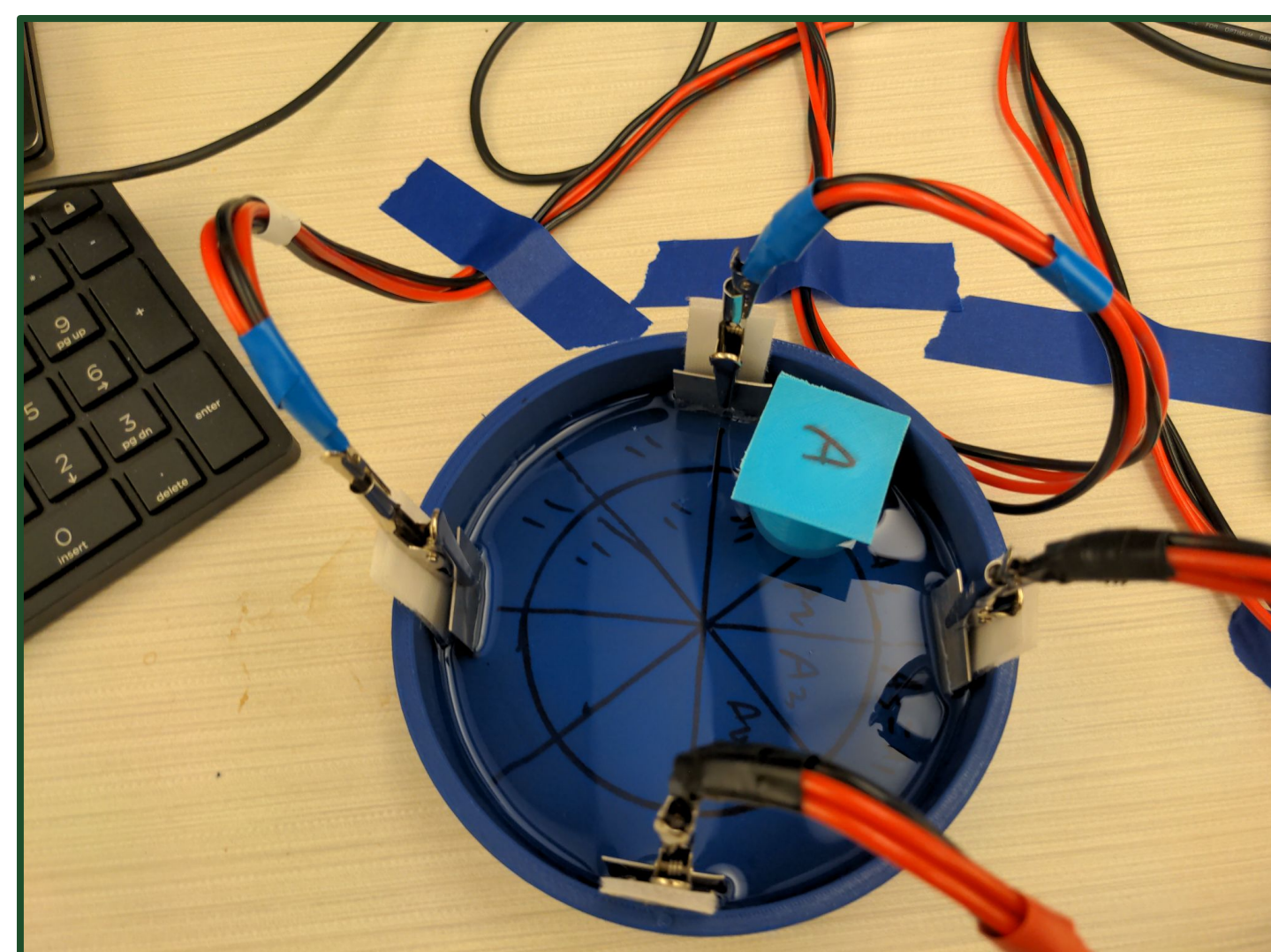


Figure H2: Test rig and electrodes with OHR in AB_2

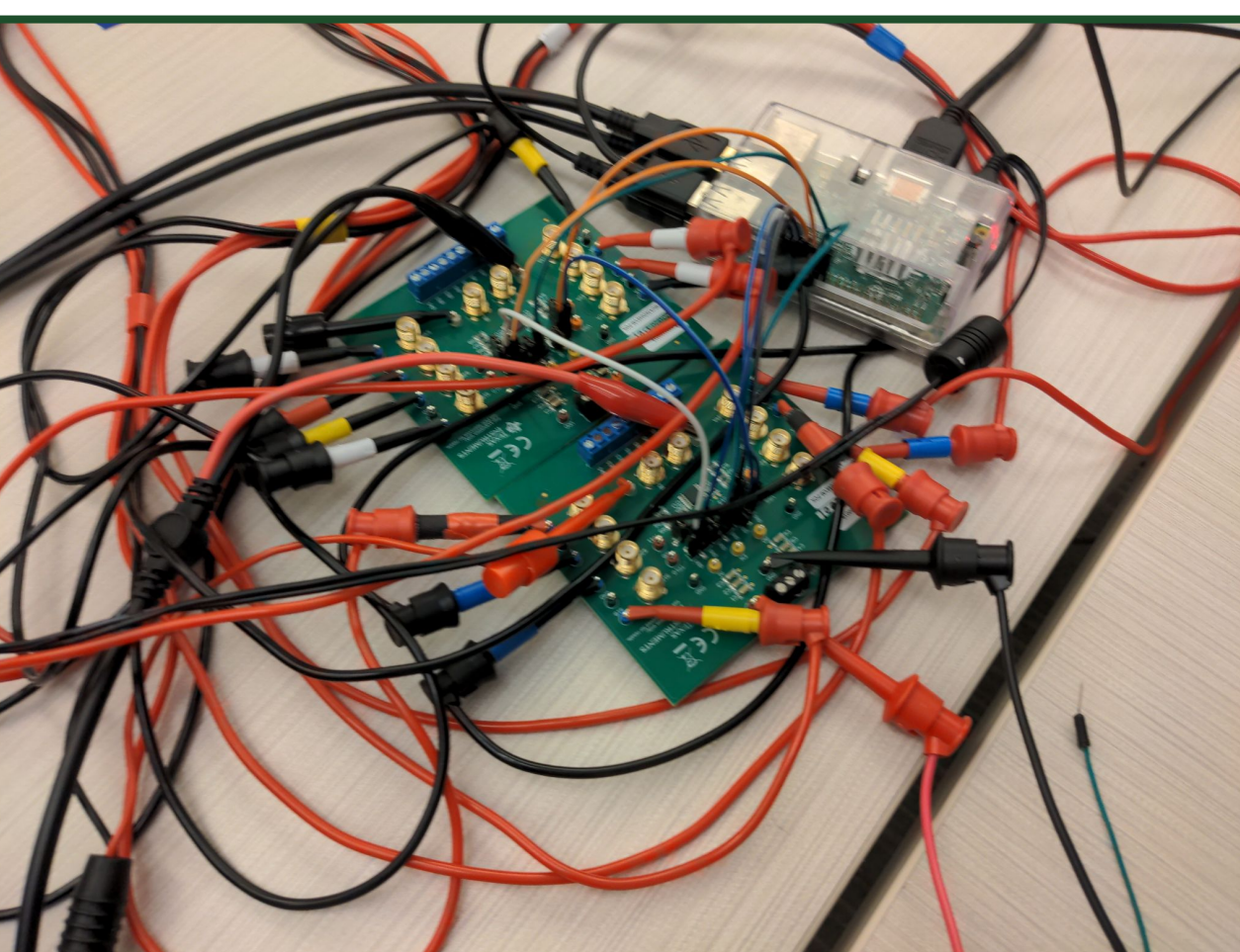


Figure H3: RPi and EVM connected to electrodes

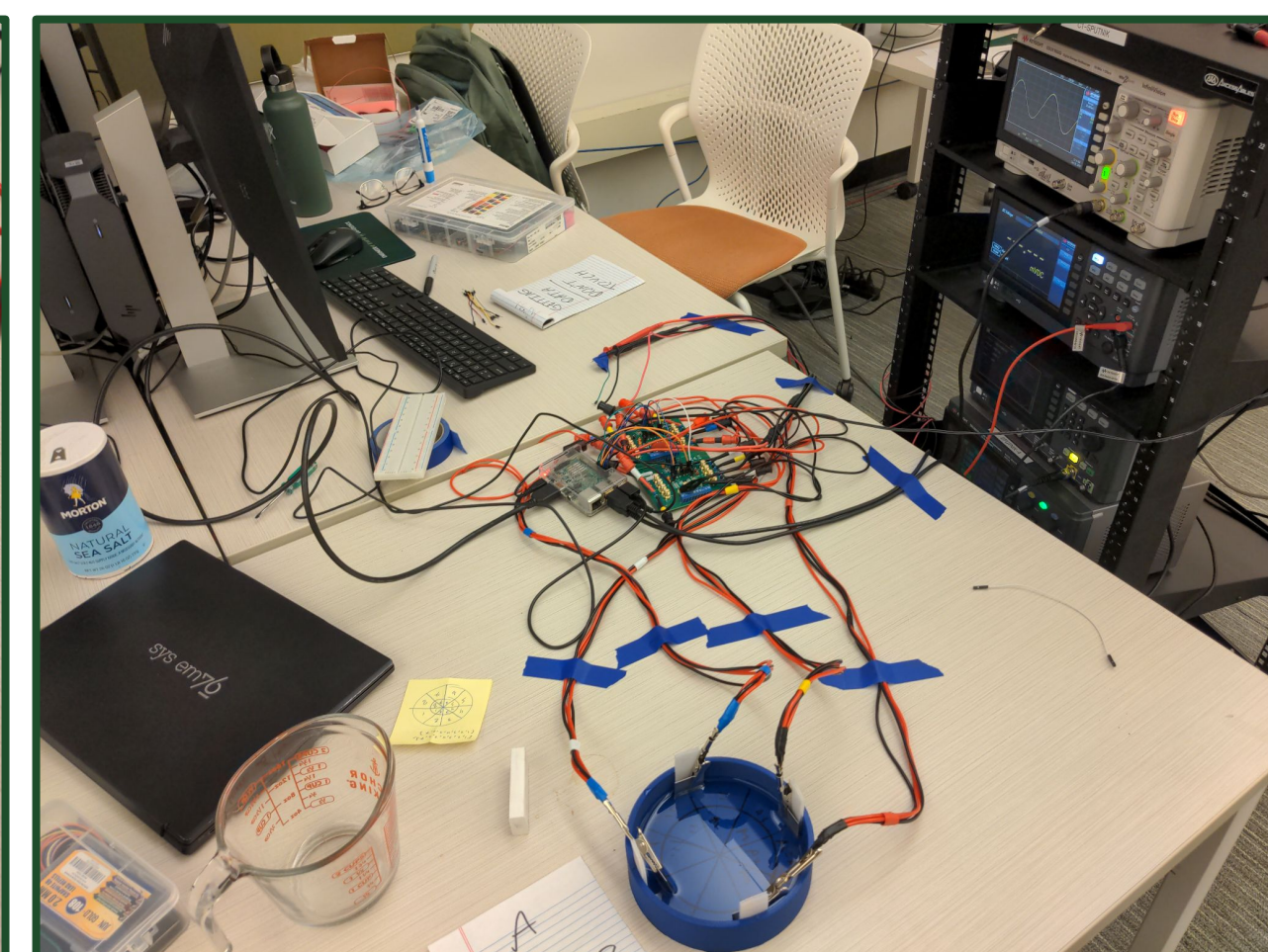


Figure H4: Full experimental setup, with oscilloscope, etc.

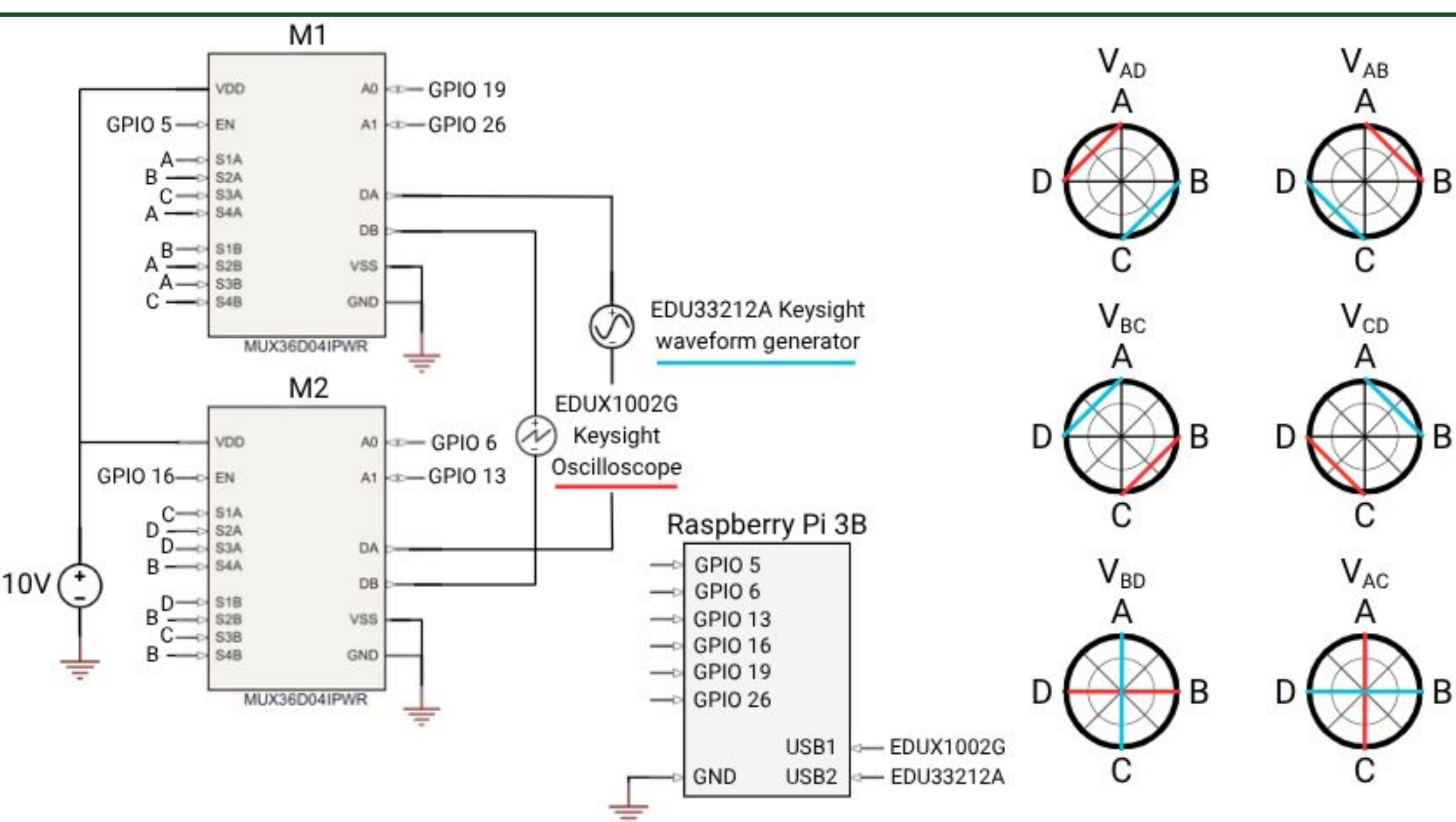


Figure H5: Full circuit diagram

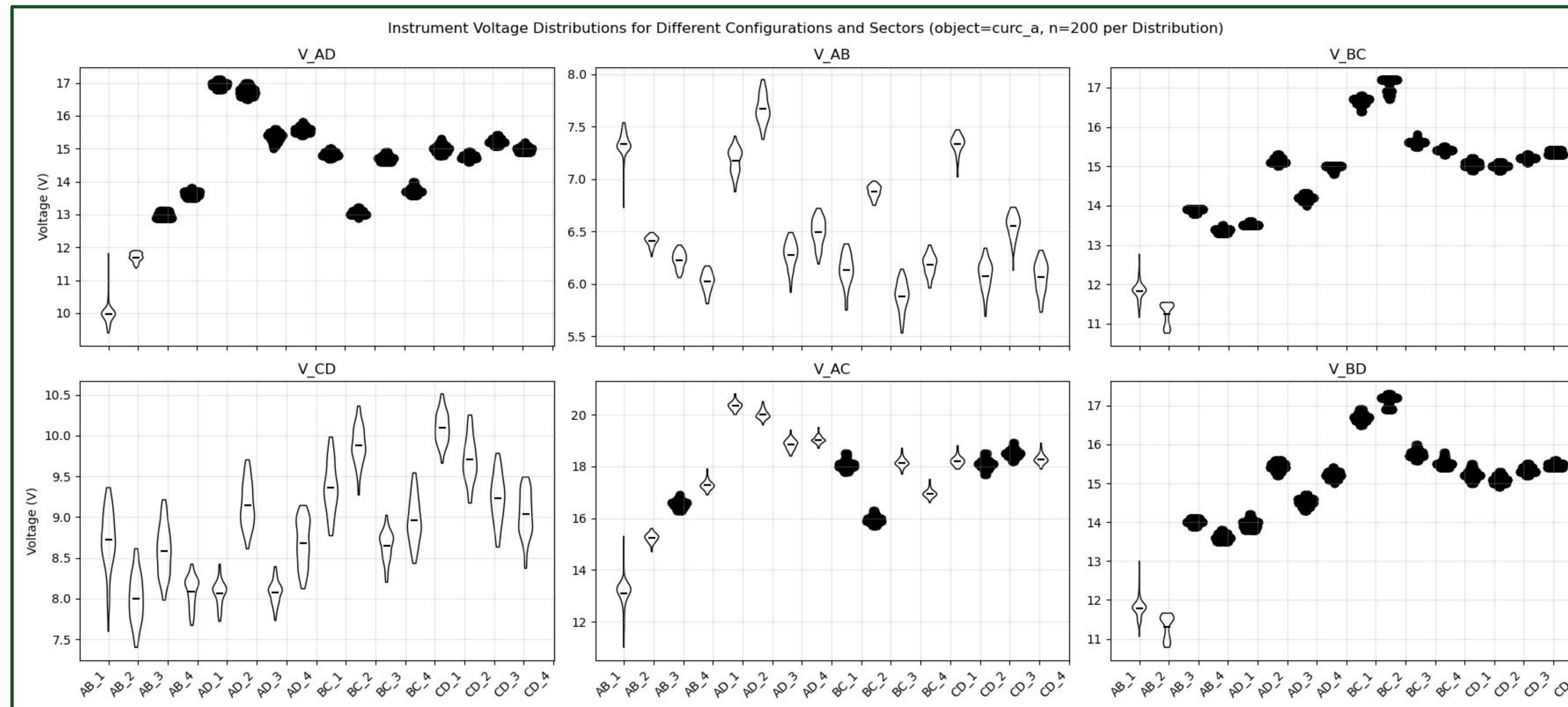


Figure D1: 15.0 mm radius OHR voltage data per sector for all 6 voltage configurations (see Figure H5)

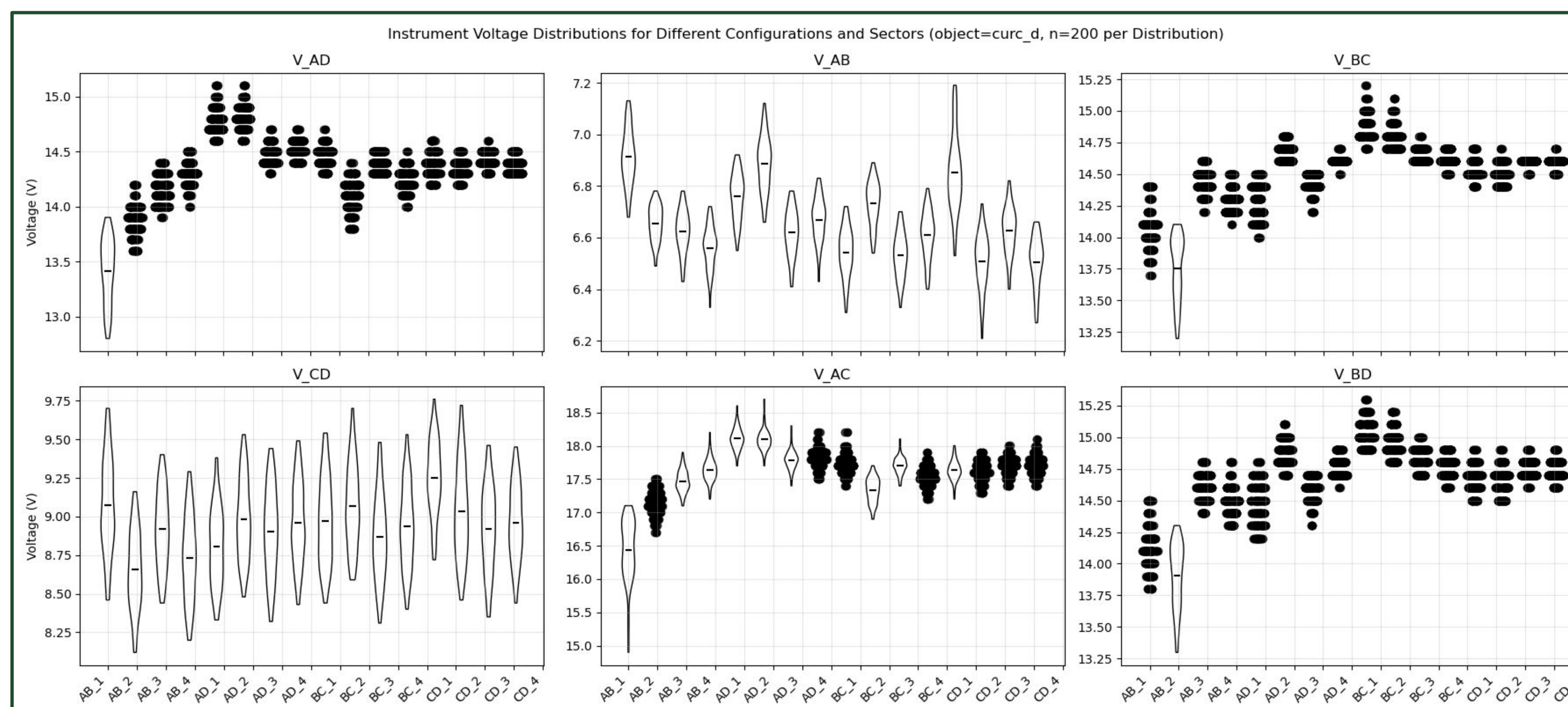


Figure D2: 6.0 mm radius OHR voltage data per sector for all 6 voltage configurations (other OHR data omitted from poster)